

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

BAUSCH & LOMB INCORPORATED,
Petitioner,

v.

ZEAVISION, LLC,
Patent Owner.

IPR2022-00089
Patent 10,307,384 B2

Before ZHENYU YANG, TINA E. HULSE, and ROBERT A. POLLOCK,
Administrative Patent Judges.

HULSE, *Administrative Patent Judge.*

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Bausch & Lomb Incorporated (“Petitioner”) filed a Petition requesting an *inter partes* review of claims 12–14, 16–21, 23, and 24 of U.S. Patent No. 10,307,384 B2 (Ex. 1001, “the ’384 patent”), owned by ZeaVision, LLC (“Patent Owner”). Paper 2 (“Pet.”). Upon considering the Petition and Patent Owner’s Preliminary Response (Paper 6), we instituted an *inter partes* review of the challenged claims of the ’384 patent. Paper 7 (“Dec. Inst.” or “Institution Decision”).

Patent Owner then filed a Response to the Petition (Paper 10, “PO Resp.”), Petitioner filed a Reply (Paper 20, “Pet. Reply”), and Patent Owner filed a Sur-reply (Paper 24, “PO Sur-reply”).

An oral hearing was held on February 3, 2023, a transcript of which has been entered in the record. Paper 32 (“Tr.”).

We have authority under 35 U.S.C. § 6. We issue this Final Written Decision under 35 U.S.C. § 318(a) and 37 C.F.R. § 42.73. For the reasons that follow, we determine Petitioner has shown by a preponderance of the evidence that claims 12–14, 16–21, 23, and 24 of the ’384 patent are unpatentable.

A. *Real Parties-in-Interest*

Petitioner identifies itself as the real party-in-interest. Pet. 3. Patent Owner identifies itself as the real party-in-interest. Paper 4, 2.

B. *Related Proceedings*

The parties state that the ’384 patent has been asserted against Petitioner in a pending case before the U.S. District Court for the Eastern District of Missouri: *ZeaVision, LLC v. Bausch & Lomb Inc.*, No. 4:21-cv-00739 (E.D. Mo.). Pet. 4; Paper 4, 2.

C. The '384 Patent

The '384 patent is entitled “Zeaxanthin Formulations with Additional Ocular-Active Nutrients, for Protecting Eye Health and Treating Eye Disorders.” Ex. 1001, code (54). The '384 patent was filed on May 25, 2018, and claims priority to a series of continuation applications, the earliest of which was filed on December 23, 2003. *Id.*, code (63). The '384 patent also claims priority to Provisional Application No. 60/453,522, which was filed on March 10, 2003.¹ *Id.*, code (60).

The '384 patent “relates to orally-ingested formulations for treating or preventing eye diseases and vision problems.” *Id.* at 1:22–24. The Specification explains that carotenoids are important because they can absorb ultraviolet (“UV”) and blue light. *Id.* at 4:25–33. Carotenoids are effective anti-oxidants that neutralize oxygen free radicals that result from UV radiation damage. *Id.* at 5:3–27. Carotenoids, however, cannot be synthesized by humans, who must therefore obtain carotenoids in food or dietary supplements. *Id.* at 4:33–35.

Zeaxanthin and lutein are the only two carotenoids found in the macula, which is the center of the retina. *Id.* at 6:27–31. The macula is responsible for fine resolution and must therefore be healthy for good vision. *Id.* at 9:16–18. The macula deposits the highest concentration of zeaxanthin in the center, most crucial part of the macula and deposits lutein in a ring around the zeaxanthin. *Id.* at 9:46–10:3. Macular degeneration is the

¹ The provisional application number on the face of the patent (and the Application Data Sheet, Ex. 1002, 11) appears to be a typographical error, and the correct application number should have been 60/435,522, which was filed December 23, 2002, as stated under “Related Applications” in the Specification. Ex. 1001, 1:14–16. We address the implications of this error below.

leading cause of blindness among the elderly. *Id.* at 9:24–25. Studies have shown that people with below-normal concentrations of zeaxanthin in their macula have higher risks and rates of macular degeneration than those with normal levels. *Id.* at 9:45–49.

As described in the '384 patent, “human retinas have developed and evolved with a notable and substantial preference for zeaxanthin, over lutein.” *Id.* at 10:31–33. Zeaxanthin, however, is scarce in dietary sources, whereas lutein is relatively abundant. *Id.* at 10:12–13. The Specification states that the only isomer of zeaxanthin that is present in dietary sources is the 3R, 3'R stereoisomer (or R—R or 3R-3R' isomer). *Id.* at 10:30–52. Although the human retina can convert lutein into zeaxanthin, it cannot convert lutein into the same isomer of zeaxanthin that is present in the natural diet (i.e., the R—R isomer). *Id.* at 10:36–40. Rather, the retina can only convert lutein into the S—R stereoisomer, or meso-zeaxanthin, which has not been shown to exist in any known dietary sources. *Id.* at 10:48–55.

Accordingly, it is an object of the invention to disclose oral formulations for mammals that “contain zeaxanthin as an essential and critical ingredient, and which also contain at least two or more other agents that have been proven . . . to act in a synergistic and potentiating manner with zeaxanthin, to provide improved efficacy in preventing or treating eye diseases.” *Id.* at 15:45–53.

D. Illustrative Claim

Petitioner challenges claims 12–14, 16–21, 23, and 24 of the '384 patent. Claim 12, the only independent claim challenged, is representative and is reproduced below.

12. A nutrient formulation for oral ingestion for protecting a user's eyes from the harmful effect of blue light, the formulation comprising:

- a. zeaxanthin that is present in the 3R-3R' stereoisomer form and is substantially free of meso-zeaxanthin stereoisomers;
- b. lutein;
- c. vitamin C; and
- d. mixed tocopherols;

wherein the nutrient formulation is provided in the form selected from the group consisting of capsules, tablets, liquids, and powders for oral ingestion.

Ex. 1001, 37:13–25.

Dependent claims 13, 14, 18–20, 23, and 24 recite additional components of the nutrient formulation; dependent claim 16 recites a specific dosage of zeaxanthin; dependent claim 17 recites a specific dosage form for the formulation; and dependent claim 21 recites specific eye disorders of the user of the formulation. *Id.* at 37:26–38:27.

E. The Asserted Grounds of Unpatentability

Petitioner asserts that claims 12–14, 16–21, 23, and 24 are unpatentable on the following grounds:

Claim(s) Challenged	35 U.S.C. § ²	Reference(s)/Basis
12, 13, 17–21, 23, 24	102(b)	Lang ³
12, 17, 18, 20	102(a), (e)	Barker ⁴
12, 13, 16–21, 23, 24	103(a)	Lang, Garnett ⁵

² Because the claims at issue have an effective filing date before March 16, 2013, the effective date of the applicable provisions of the Leahy Smith America Invents Act, Pub. L. No. 112–29, 125 Stat. 284 (2011) (“AIA”), we apply the pre-AIA versions of 35 U.S.C. §§ 102 and 103(a) in this Decision.

³ WO 01/19383 A1, published Mar. 22, 2001 (“Lang,” Ex. 1009).

⁴ WO 03/063848 A1, published Aug. 7, 2003 (“Barker,” Ex. 1011).

⁵ Garnett et al., US 5,747,544, issued May 5, 1998 (“Garnett,” Ex. 1012)

Claim(s) Challenged	35 U.S.C. § ²	Reference(s)/Basis
14	103(a)	Lang, Garnett, Brown ^{6, 7}
12, 13, 17–21, 23, 24	103(a)	Lang, Roche ⁸
14	103(a)	Lang, Roche, Brown

Petitioner also relies on the Declaration of Dr. Elizabeth J. Johnson.
Ex. 1024. Patent Owner relies on the Declaration of Dr. C. Kathleen Dorey.
Ex. 2001.

II. ANALYSIS

A. *Person of Ordinary Skill in the Art*

In determining the level of ordinary skill in the art, we consider the type of problems encountered in the art, the prior art solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Petitioner asserts that a person of ordinary skill in the art (“POSITA”) at the time of the invention would have had:

a Masters or a Ph.D. degree in nutritional science, chemistry, biology, biochemistry, or a related discipline or an M.D. or doctor of optometry (“O.D.”), and a few years of practical experience in the nutritional supplement area, including an

⁶ Brown et al., *Nutrition Supplements and the Eye*, 12 EYE 127–33 (1998) (“Brown,” Ex. 1014).

⁷ Although Petitioner groups Brown together with Lang and Garnett and Lang and Roche in Grounds 3 and 4, respectively, Petitioner only relies on Brown for claim 14. *See* Pet. 6, 56, 72–73. We have therefore separated the challenge to claim 14 from that of the other claims, which only rely on the combination of Lang and Garnett and of Lang and Roche.

⁸ FDA Docket No. 95S-0316, 75-Day Premarket Notification for New Dietary Ingredient Zeaxanthin by Roche Vitamins, Inc. (opened for public inspection June 2001) (“Roche,” Ex. 1020).

understanding of formulations that could be considered for promoting eye health, including preventing or treating eye disorders such as macular degeneration. Alternatively, such a person could have been a person with a Bachelor's degree with a greater number of years of experience. The POSITA would have, or would have access to, general information regarding eye health and eye disorders, the convenience and needs of patients who suffer from eye disorders such as macular degeneration, and background information. And to the extent needed, the POSITA would have collaborated with others having ordinary skill in the areas pertinent to the subject matter of the '384 Patent.

Pet. 7–8 (citing Ex. 1024 ¶ 16).

In our Institution Decision, we adopted Petitioner's definition (Dec. Inst. 6–7), which Patent Owner states that it “does not contest.” PO Resp. 17.

Because the parties do not dispute the level of ordinary skill in the art, we adopt Petitioner's definition as consistent with the prior art's demonstration of the level of ordinary skill in the art at the time of the invention. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required “where the prior art itself reflects an appropriate level and a need for testimony is not shown” (*quoting Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985))). Moreover, we find that Dr. Johnson and Dr. Dorey are both qualified to opine from the perspective of a skilled artisan as both are persons of at least ordinary skill in the art. *See* Ex. 1024 ¶¶ 2–11, App'x A; Ex. 2001 ¶¶ 4–14, App'x A; *see also Kyocera Senco Indus. Tools Inc. v. Int'l Trade Comm'n*, 22 F. 4th 1369, 1376–77 (Fed. Cir. 2022) (“To offer expert testimony from the perspective of a skilled artisan in a patent case—like for claim construction, validity, or infringement—a witness must at least have ordinary skill in the art.”).

B. Claim Construction

In an *inter partes* review, the Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 100(b) (2021). Under that standard, claim terms “are generally given their ordinary and customary meaning” as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc). “In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence.” *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17). Extrinsic evidence is “less significant than the intrinsic record in determining ‘the legally operative meaning of claim language.’” *Phillips*, 415 F.3d at 1317.

1. Preamble

Petitioner contends the preamble of claim 12 is limiting. Pet. 16. Petitioner asserts that the preamble provides antecedent basis for the term “the nutrient formulation” recited in the body of the claim. *Id.* Moreover, Petitioner asserts the specification repeatedly describes the use of the formulation to protect the eyes against damage from UV and blue light. *Id.* at 16–17. Petitioner also argues that Patent Owner amended the claims to recite the preamble language in response to prior art rejections. *Id.* at 17. Thus, according to Petitioner, a person of ordinary skill in the art would have understood the preamble of the claim to limit the scope of the claim.

In our Institution Decision, we preliminarily determined the preamble is limiting, noting the preamble provides antecedent basis for “the nutrient

formulation” in the body of claim 12 and “the user” in dependent claim 22. Dec. Inst. 8–9. We also found the Specification consistently describes the benefits of carotenoids, such as zeaxanthin and lutein, as including the ability to absorb UV and blue light. *Id.* at 9. Because we found it was a “close case,” we invited the parties to address this issue further during trial. Dec. Inst. 7–9.

In its Response, Patent Owner disagrees with our construction, contending that the preamble of claim 12 is not limiting. PO Resp. 13–17. Patent Owner asserts the invention is structurally complete without the preamble, and “[t]he preamble describes an intended use of the formulation, namely protecting a user’s eyes from the harmful effect of blue light.” *Id.* at 15. During the hearing, however, Patent Owner reversed its position and conceded that the preamble is limiting:

JUDGE HULSE: So then are you saying that [the preamble] is limiting?

MR. SIMON: Yes. We’re conceding.

Tr. 45:18–46:3.

Although there is no longer a dispute, we agree with the parties that the preamble is limiting in this case for the reasons stated in our Institution Decision. Dec. Inst. 8–9. We note the Federal Circuit’s decision in *Proveris Scientific Corp. v Innovasystems, Inc.*, 739 F.3d 1367 (Fed. Cir. 2014), is instructive and consistent with our determination. In that case, the claim was directed to:

3. An apparatus for producing image data representative of at least one sequential set of images of a spray plume . . . comprising:

. . .

an imaging device for generating the image data representative of an interaction between the illumination and the spray plume along the at least one geometric plane.

Id. at 1372. The Federal Circuit held the preamble to the apparatus claim is limiting because “the specification identifies the invention as producing a ‘sequential set of images’ and focuses on the ability of the invention to capture ‘the time evolution of the spray.’” *Id.* at 1373. The court noted that the preamble was the only reference in any of the independent claims to that inventive concept. Accordingly, the court held “[t]his fact alone is likely sufficient to support a conclusion that the preamble is limiting.” *Id.* As further support, the court noted that the preamble provides antecedent basis for “the image data” in the claim body, which “is defined in greater detail in the preamble.” *Id.*

Similarly, here, the Specification consistently states that the invention relates to oral formulations for protecting eye health and that the benefits of carotenoids, such as zeaxanthin and lutein, include the ability to absorb UV and blue light. *See, e.g.,* Ex. 1001, Abstract (“Oral formulations for promoting eye health”), 4:25–31 (stating carotenoids such as zeaxanthin are “very good at absorbing ultraviolet (UV) and blue light”), 15:45–47 (“one object of this invention is to disclose multi-component orally-ingestible formulations for protecting eye health in mammals”). Like the preamble in *Proveris*, the preamble in claim 12 is the only reference in any of the independent claims to the allegedly inventive concept of protecting eye health from the harmful effects of blue light. *Proveris*, 739 F.3d at 1373.

Moreover, the preamble of claim 12 recites: “A nutrient formulation for oral ingestion for protecting a user’s eyes from the harmful effect of blue light, the formulation comprising:.” Ex. 1001, 37:13–15. The body of claim

12 recites “wherein *the* nutrient formulation is provided in the form selected from the group consisting of capsules, tablets, liquids, and powders for oral ingestion.” Ex. 1001, 37:22–24 (emphasis added). Similarly, claim 22, which depends from claim 12, relies on the preamble of claim 12 for antecedent basis for the limitation “wherein *the* user has macular degeneration and a lens-related disorder.” *Id.* at 38:21 (emphasis added). Thus, “the nutrient formulation” and “the user” in the claim body derives antecedent basis from the “nutrient formulation” and “user” that is “defined in greater detail in the preamble.” *See Proveris*, 739 F.3d at 1373. Accordingly, reading the patent as a whole, because the inventor clearly relied on both the preamble and the body of claim 12 to define the claimed invention, the preamble of claim 12 should be construed as limiting. *See id.*

2. *Remaining Claim Terms*

Petitioner contends the term “substantially-free of meso-zeaxanthin stereoisomers” should have its plain and ordinary meaning, and a person of ordinary skill in the art would understand this term to mean that the zeaxanthin that is present in the 3R-3R' stereoisomer form is “largely but not wholly free of meso-zeaxanthin stereoisomers.” Pet. 19 (citing Ex. 1024 ¶ 57, Ex. 1031, 3).

Petitioner also contends the term “mixed tocopherols” should be construed to mean “a mixture of tocopherols such as alpha-, beta-, gamma-, and delta-tocopherols.” Pet. 22–24 (citing Ex. 1024 ¶¶ 65–66).

Patent Owner does not address either of these claim terms in its Response. *See generally* PO Resp.

Because there does not appear to be any dispute over the construction of these terms, we adopt Petitioner’s proposed constructions as they are consistent with the plain and ordinary meaning of the terms. *See Wellman*,

Inc. v. Eastman Chem. Co., 642 F.3d 1355, 1361 (Fed. Cir. 2011) (“[C]laim terms need only be construed ‘to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))).

C. Anticipation over Lang

Petitioner asserts that claims 12, 13, 17–21, 23, and 24 are unpatentable as anticipated by Lang. Pet. 24–40. Patent Owner opposes Petitioner’s assertion. PO Resp. 19–26. Having considered the evidence and arguments presented at trial, we determine that Petitioner has established by a preponderance of evidence that claims 12, 13, 17–20, 23, and 24 are anticipated by Lang, but not claim 21.

1. Lang (Ex. 1009)

Lang is a published PCT application entitled “Stable Carotene-Xanthophyll Beadlet Compositions and Methods of Use.” Ex. 1009, code (54). Lang was published on March 22, 2001, thereby making it prior art under 35 U.S.C. § 102(b), which Patent Owner does not dispute. *See generally* PO Resp.

Lang relates to beadlet formulations for including in dietary supplements comprising “one or more xanthophylls; one or more carotenes, retinoids, or combinations thereof; one or more antioxidants; and excipients.” Ex. 1009, 4:7–10. Lang discloses that the dietary supplements are used for “the enhancement of ocular health and the prophylaxis of retinal disorders, including age-related macular degeneration.” *Id.* at 4:17–18.

Lang further states that examples of xanthophylls include lutein and zeaxanthin. *Id.* at 8:7–9. Lang states that “[i]t has been suggested that the particular xanthophylls, zeaxanthin and its isomer lutein, may be beneficial in improving the health of the macula and the clarity of the lens.” *Id.*

at 2:17–19. Lang teaches that xanthophylls may be obtained from various dietary sources, such as corn and leafy green vegetables. *Id.* at 8:3–4. Lang also teaches that examples of natural antioxidants and related derivatives include alpha-, beta-, gamma-, delta- and epsilon-tocopherol, as well as Vitamin C. *Id.* at 9:23–10:2.

An example of a beadlet composition is shown in Example 3, reproduced below:

Example 3

Ingredient	Amount (% w/w)
Beta Carotene (natural)	3
Lutein	1.5
Zeaxanthin	1.5
Hydrolyzed Gelatin	30
Vegetable Oil/Oleoresin	23
Sucrose	14.6
Food Grade Corn Starch	17
Ascorbyl Palmitate	2.7
Tocopherol(s)	1.7
Water	5

Example 3 shows a beadlet composition including lutein, zeaxanthin, and tocopherol(s). *Id.* at 12:19. Lang also provides examples of dietary supplement gelatin capsules that incorporate beadlets, like that of Example 3, along with other nutrients such as Vitamin E to treat various ocular diseases. *See, e.g., id.* at 23:1–8 (Example 12).

2. Analysis

Anticipation requires that “each and every element as set forth in the claim is found, either expressly or inherently described, in a single prior art

reference.” *In re Robertson*, 169 F.3d 743, 745 (Fed. Cir. 1999) (citation omitted). “To establish inherency, the extrinsic evidence ‘must make clear that the missing descriptive matter is necessarily present in the thing described in the reference, and that it would be so recognized by persons of ordinary skill.’” *Id.* (citation omitted). Moreover, to anticipate, a prior art reference must “disclose[] within the four corners of the document not only all of the limitations claimed but also all of the limitations arranged or combined in the same way as recited in the claim.” *Net MoneyIN, Inc. v. Verisign, Inc.*, 545 F.3d 1359, 1371 (Fed. Cir. 2008).

a) Claim 12

Having considered the arguments and evidence presented at trial, we are persuaded that Petitioner has shown by a preponderance of the evidence that Lang discloses each limitation of claim 12. *See* Pet. 25–33. We find Lang discloses the preamble because it teaches the use of dietary supplements that include lutein and zeaxanthin, which are known to absorb photoexcitative radiation of short visible wavelength that can damage the eyes. Ex. 1009, 2:16–3:6, 4:16–19; Ex. 1015, 1455S–56S, Abstract; Ex. 1024 ¶¶ 81–82. Lang also discloses the 3R-3R' form of zeaxanthin and is substantially free of meso-zeaxanthin because Lang teaches obtaining zeaxanthin from corn, and a person of ordinary skill in the art would understand that the 3R-3R' stereoisomer is the only zeaxanthin present in dietary sources. Ex. 1009, 8:3–7, 7:24; Ex. 1024 ¶ 84; Ex. 1001, 10:40–43 (stating “[t]he only isomer of zeaxanthin that is present in dietary sources is the 3R, 3'R stereoisomer”). Finally, Example 12 of Lang, which incorporates Example 3 (referred to by Petitioner as “Lang 12-3”), discloses lutein, vitamin C, and mixed tocopherols in the form of gelatin capsules for oral ingestion. Ex. 1009, Example 12; Ex. 1024 ¶¶ 87–91.

Patent Owner asserts that Lang does not disclose “mixed tocopherols” or the use of zeaxanthin present in the 3R-3R' stereoisomer form. PO Resp. 19–24. We disagree and address each of those arguments below.

(1) *mixed tocopherols*

Patent Owner argues that Lang does not disclose “mixed tocopherols,” as required by claim 12. *Id.* at 19–20. Patent Owner argues that “[a]lthough Lang discloses multiple tocopherols, it fails to disclose mixing those tocopherols as required by the ’384 Patent.” *Id.* at 19. Patent Owner asserts that a person of ordinary skill in the art would not have understood Lang to require mixed tocopherols because “the benefit of multiple tocopherols in the same supplement was not yet clear in the art.” *Id.* at 20 (citing Ex. 2002 ¶ 27). According to Patent Owner, a POSITA would not have had a reason to combine more than one tocopherol in Lang because the tocopherols in the beadlets were added to prevent degradation of carotenoid components, which only requires one tocopherol. *Id.* (citing Ex. 2002 ¶ 27).

Moreover, Patent Owner asserts that a POSITA would not have recognized the importance of mixed tocopherols at the time of invention, because alpha-tocopherol was the “star of the show” in Vitamin E research and was the form included in multivitamins and dietary supplements. *Id.* at 21 (citing Ex. 2002 ¶ 30). According to Patent Owner, gamma-tocopherol was known only for treating inflammation caused by cell damage; and beta- and delta-tocopherols had not been researched. *Id.* at 22 (citing Ex. 2002 ¶ 33). A POSITA, therefore, would not have recognized the benefit of including other tocopherols in the formulation to protect the retina and lens. *Id.* at 21–22 (citing Ex. 2002 ¶ 36). As such, Patent Owner asserts that because a POSITA would not have had a reason to mix tocopherols in a

supplement, Lang does not enable a POSITA to make the invention without undue experimentation. *Id.* at 22 (citing Ex. 2002 ¶¶ 29, 30, 33).

We are not persuaded by Patent Owner’s arguments, and we find Petitioner has shown by a preponderance of the evidence that Lang discloses mixed tocopherols. Example 3 of Lang discloses “tocopherol(s)” as an ingredient of the beadlet, indicating the formulation expressly includes a plurality of tocopherols that are mixed within the beadlet. *See* Ex. 1009, 14. Moreover, Example 12, which incorporates the beadlet of Example 3, discloses Vitamin E among “Other nutrients” in addition to the composition of the beadlet. *Id.* at 23. Thus, at a minimum, Lang 12-3 teaches a formulation that includes multiple tocopherols and Vitamin E.

The parties dispute what isoforms a person of ordinary skill in the art would understand Vitamin E to include. *See* Pet. 32; PO Resp. 19. As an initial matter, we note that Lang teaches that Vitamin E is an example of a “natural antioxidant” that includes “alpha-, beta-, gamma-, delta- and epsilon-tocopherol.” Ex. 1009, 9–10. Consistent with Lang’s disclosure, Petitioner’s expert, Dr. Johnson, asserts that Vitamin E from natural sources was known to encompass alpha-, beta-, gamma-, and delta-tocopherols. *See* Ex. 1024 ¶ 90. As support, Dr. Johnson cites various references stating the same. *Id.* (citing Ex. 1035, Abstract, 44 (stating “[n]atural vitamin E includes four tocopherols (α , β , γ , δ)”); Ex. 1022, 3 (defining Vitamin E as a “[g]eneral term referring to 2 groups of closely related 6-chromanol derivatives known as tocopherols and tocotrienols, each comprised of four homologs (α , β , γ , and δ)”). Patent Owner’s expert, Dr. Dorey, on the other hand, asserts that Vitamin E refers to alpha-tocopherol alone. Ex. 2001 ¶ 18. Unlike Dr. Johnson, however, Dr. Dorey cites no objective evidence in support of her opinion. Our rules are clear that “[e]xpert testimony that does

not disclose the underlying facts or data on which the opinion is based is entitled to little or no weight.” *See* 37 C.F.R. § 42.65(a). As such, we credit the testimony of Dr. Johnson over Dr. Dorey and find that Lang’s disclosure of Vitamin E in the nutrient formulation discloses “mixed tocopherols” of the claims.

We also find that Lang’s use of the plural “tocopherol(s)” in Example 3 (and therefore in the gelatin capsule of Example 12) is sufficient to disclose two or more tocopherols mixed in one supplement, as required by the claims.⁹ That Lang uses “(s)” to indicate that the beadlet can have a single tocopherol or multiple tocopherols does not negate the express disclosure of multiple tocopherols. We are also not persuaded that the claim requires Lang to disclose a mixing step of the tocopherols, as Patent Owner asserts. PO Resp. 19. The claim is a composition claim to a nutrient formulation, not a method claim. And, as Patent Owner admits, a “POSITA would understand ‘mixed tocopherols’ as a formulation including two or more tocopherols.” *Id.* at 19–20. No mixing step is required for the claim.

Patent Owner repeatedly argues that “Lang teaches one tocopherol is sufficient” and “there would be no motivation to mix tocopherols” because the tocopherols are only disclosed as a preservative or for inhibiting degradation of the carotenoid products. *Id.* at 22, 23. But this is an anticipation ground. There is no motivation required when the limitations of

⁹ Consistent with this finding, we also note that claim 4 of Lang recites: “A beadlet according to Claim 1, wherein the antioxidant(s) is/are selected from the group consisting of: . . . , alpha-, beta-, gamma-, delta- and epsilon-tocopherol. . . .” Ex. 1009, 25 (claim 4). This claim also suggests a beadlet may comprise multiple isoforms of tocopherol.

the claim are expressly or inherently disclosed in the reference, as they are in Lang.

Patent Owner's enablement argument similarly falls short. Patent Owner seems to assert that because a POSITA would not have had a reason to mix tocopherols in a supplement, Lang is not enabled. *Id.* at 22. We disagree. Lang discloses a dietary gelatin capsule that incorporates specific amounts of various ingredients, including tocopherols, into the beadlets of Example 3, with specific amounts of other nutrients, like Vitamin E, in Example 12. Ex. 1009, 14 (Example 3), 23 (Example 12). Given this detailed disclosure, we are not persuaded—nor does Patent Owner argue—that a POSITA would not know how to make or use such an oral formulation.

Accordingly, we are not persuaded that Lang fails to disclose “mixed tocopherols.”

(2) *zeaxanthin present in 3R-3R' stereoisomer form and is substantially free of meso-zeaxanthin stereoisomers*

Patent Owner also argues that Lang does not disclose the use of zeaxanthin in the 3R-3R' stereoisomer form and that a POSITA would not understand Lang to require 3R-3R' zeaxanthin that is substantially free of meso-zeaxanthin. PO Resp. 23–24 (citing Ex. 2002 ¶¶ 50–52). Patent Owner further argues that a POSITA “would not understand Lang to teach that increasing a user's intake of exclusively 3R-3R' and not other zeaxanthin stereoisomers supports the health of the macula and the clarity of the lens.” *Id.* at 24 (citing Ex. 2002 ¶ 53).

We are not persuaded. Although Lang may not use the term “3R-3R' zeaxanthin,” Lang expressly discloses that zeaxanthin “may be obtained from various sources such as vegetables and herbal components, such as

corn, leafy green vegetables, and marigolds.” Ex. 1009, 8. As Dr. Johnson explains, “the 3R-3R' stereoisomer is the only zeaxanthin present in dietary sources such as corn.” Ex. 1024 ¶ 84 (citing Ex. 1016, 122 Table 1 (showing corn contains >99% 3R-3R' zeaxanthin)). Even the '384 patent Specification is consistent with Dr. Johnson's testimony, stating “[t]he only isomer of zeaxanthin that is present in dietary sources is the 3R, 3'R stereoisomer.” Ex. 1001, 10:40–42. We, therefore, find that Lang discloses the use of zeaxanthin in Example 12–3 and that the zeaxanthin obtained from dietary sources is exclusively 3R-3R' zeaxanthin, which is substantially free of meso-zeaxanthin. *See* Ex. 1009, 8, 14, 23. That is sufficient for anticipation. We, therefore, find that Lang discloses the 3R-3R' stereoisomer that is substantially free of meso-zeaxanthin stereoisomers, as required by claim 12.

Accordingly, having considered the arguments and evidence presented at trial, we determine that Petitioner has shown by a preponderance of the evidence that Lang anticipates claim 12.

b) Dependent Claims

Regarding claims 13, 17–20, and 24, which depend from claim 12, Petitioner relies on Example 12 and provides a detailed claim chart identifying how the Example discloses the limitations of each claim. Pet. 33–35 (citing Ex. 1009, Example 12, Example 3; Ex. 1024 ¶ 93). For claim 23, Petitioner relies on Example 10, which incorporates the beadlet of Example 3, to disclose the additional limitation of taurine. Pet. 37–40 (citing Ex. 1009, Example 10, Example 3; Ex. 1024 ¶ 101). We are persuaded that Petitioner has shown by a preponderance of the evidence that Lang discloses each limitation of those claims, for the reasons stated by

Petitioner. *See* Pet. 33–35, 37–40; *see also* Ex. 1009, Examples 3, 10, 12; Ex. 1024 ¶¶ 93, 101.

In response, Patent Owner relies on its arguments for claim 12 for dependent claims 13, 18, 19, 23, and 24. PO Resp. 24–26. Because, as explained above, we do not find Patent Owner’s arguments persuasive, we determine, based on the evidence and argument presented at trial, that Petitioner has shown by a preponderance of the evidence that claims 13, 18, 19, 23, and 24 are anticipated by Lang.

Regarding dependent claim 17, Patent Owner argues Lang does not disclose “capsules containing oily carrier materials,” because Lang teaches the use of oils within the beadlets. PO Resp. 24–25. We are not persuaded. The gelatin capsule of Example 12 comprises beadlets that contain vegetable oil/oleoresin. Ex. 1009, Example 3, Example 12. Thus, the “capsules contain[] oily carrier materials,” according to the claim, because the beadlets—which are within the capsules—contain oily carrier materials.

We are not persuaded that the ’384 patent “explicitly disclaims this location,” as asserted by Patent Owner. PO Resp. 25 (citing Ex. 2002 ¶ 41). Dr. Dorsey quotes the portion of the ’384 patent that states, “If desired, such oily carriers can also be formulated to carry microencapsulated beadlets or other preparations, which can contain water-soluble nutrients or any other components that are easier to handle if isolated or otherwise coated in that manner.” Ex. 1001, 20:62–67. It is unclear how this language “explicitly disclaims” the use of an oily carrier in the beadlets, as Patent Owner asserts. Dr. Dorsey asserts that the beadlets in Lang “contain oil and are surrounded by water soluble substances that could be in a capsule. In contrast, the beadlets in the ’384 Patent contain water soluble substances and are surrounded by oil in a capsule.” Ex. 2002 ¶ 41. We disagree with Dr.

Dorsey’s interpretation of the Specification as limiting the invention to such beadlets. At best, this language describes an additional embodiment that “can also be formulated” “[i]f desired,” and does not rise to the level of an “expression[] of manifest exclusion or restriction, representing a clear disavowal of claim scope.” *Aventis Pharma S.A. v. Hospira, Inc.*, 675 F.3d 1324, 1330 (Fed. Cir. 2012) (citation omitted). Thus, based on the arguments and evidence presented at trial, we determine Petitioner has shown by a preponderance of the evidence that Lang anticipates claim 17.

Finally, regarding claim 21, which requires that “the user has macular degeneration and a lens-related disorder,” Petitioner asserts that Lang discloses this limitation because Lang discloses the use of its dietary supplements to treat macular degeneration and that xanthophylls may be beneficial for improving the health of the macula and clarity of the lens. Pet. 36 (citing Ex. 1001, 2:17–19, 4:16–18, 24:9–1). As explained in our Institution Decision, we are not persuaded that this disclosure in Lang discloses the use of the formulation on a user who suffers from *both* macular degeneration and a lens-related disorder. *See* Dec. Inst. 15–16. Petitioner does not address claim 21 in its Reply. Thus, we are not persuaded that Petitioner has shown by a preponderance of the evidence that Lang anticipates claim 21.

3. Conclusion

Having considered the arguments and evidence presented at trial, we determine Petitioner has shown by a preponderance of the evidence that claims 12, 13, 17–20, 23, and 24 are anticipated by Lang, but not claim 21.

D. Anticipation over Barker

Petitioner asserts that claims 12, 17, 18, and 20 are anticipated by Barker.¹⁰ Pet. 40–45. Patent Owner opposes Petitioner’s assertion. PO Resp. 7–13, 26–28. Having considered the arguments and evidence presented at trial, we determine that Petitioner has established by a preponderance of the evidence that claims 12, 17, 18, and 20 are anticipated by Barker.

1. Barker (Ex. 1011)

Barker is a published PCT Application entitled “Lutein/Zeaxanthin for Glare Protection.” Ex. 1011, code (54). Barker was published on August 7, 2003. *Id.*, code (43). Barker is, therefore, prior art under 35 U.S.C. § 102(a). Barker was filed on January 23, 2003, designating the United States and published in English, thereby making it prior art under 35 U.S.C. § 102(e), as well. Ex. 1011, codes (22), (81), (26).

Barker relates to “administration of a colorant that is capable of being incorporated into eye tissue and/or causing yellowing of eye tissue[, which] results in an improvement of visual performance.” Ex. 1011, 1:10–12. Barker explains that such colorants, when deposited in the retina, provide a yellow filter that absorbs potentially damaging blue light. *Id.* at 1:12–15. As examples of colorants that can be used, Barker lists carotenoids such as lutein, zeaxanthin, and meso-zeaxanthin, and compounds having Vitamin A activity. *Id.* at 1:21–26. Barker also states that the colorants may be used in

¹⁰ On page 40 of its Petition, Petitioner contends that “claims 12, 16–21, and 24” are unpatentable as anticipated by Barker. Pet. 40. However, Petitioner’s arguments on pages 41–45 of its Petition appear to be directed to claims 12, 17, 18, and 20. *See* Pet. 41–45. Petitioner’s Reply confirms that the challenge is to claims 12, 17, 18, and 20. Pet. Reply 24.

combination with an antioxidant such as Vitamin E, Vitamin C, and zinc.
Id. at 1:31–2:5.

An example of a soft gelatin capsule according to the invention is shown in Example 6, reproduced below.

Ingredient	<u>Example 6</u>
	Amount per Capsule
Lutein	6 mg
Zeaxanthin	6 mg
β-Carotene	6 mg
Vitamin E (α-d,l-tocopherol)	200 mg
Vitamin C	500 mg
Zinc (as orotate)	7.5 mg
Lecithin	50 mg
Soy bean oil	200 mg

Example 6 shows a soft gelatin capsule with ingredients including lutein, zeaxanthin, Vitamin E, and soybean oil. *Id.* at 6:2.

2. Analysis

a) Priority Date for the Challenged Claims

As an initial matter, Patent Owner contends that Barker is not prior art because the '384 patent is entitled to claim priority to the December 23, 2002, filing date of its provisional application, U.S. Provisional Application No. 60/435,522 (“the '522 Provisional”). PO Resp. 7–12. Petitioner argues that Patent Owner has failed to meet its burden to claim the benefit of the '522 Provisional's filing date. Pet. Reply 2–14. We agree with Petitioner.

First, Patent Owner asserts that the '384 patent's reference to U.S. Provisional Application No. 60/453,522 (“the '53 Application”) instead of Provisional Application No. 60/435,522 on the cover of the '384 patent

(Ex. 1001, code (60)) was a transcription error that has been corrected in subsequent filings in the same family. PO Resp. 7. Petitioner, on the other hand, asserts that the “transcription error” dooms the ’384 patent’s priority claim, as the Application Data Sheet controls domestic benefit. Pet. Reply 2 (citing Ex. 1001, Cover; Ex. 1002, 11, 78; 37 C.F.R. § 1.76(d)(1)(i); MPEP § 601.05(a)).

We agree with Petitioner. 37 C.F.R. § 1.76(d)(1)(i) states:

For inconsistencies between information that is supplied by both an application data sheet under this section and other documents . . . [t]he most recent application data sheet will govern with respect to . . . domestic benefit claims.

Here, the Application Data Sheet of the ’384 patent application identifies the ’53 Application as the earliest provisional application to which the ’384 patent claims priority. Ex. 1002, 11. Although it does appear to be a typographical error, we are not permitted to correct such an error during an *inter partes* review proceeding. “Patentees may petition the Director to issue a Certificate of Correction to correct in an issued patent ‘a mistake of a clerical or typographical nature, or of minor character, which was not the fault of the Patent and Trademark Office.’” *Honeywell Int’l Inc. v. Arkema Inc.*, 939 F.3d 1345, 1348 (Fed. Cir. 2019) (quoting 35 U.S.C. § 255). “When a patentee seeks to correct a patent that is subject to a post-issuance review proceeding, the patentee must first file a motion with the Board seeking authorization to petition the Director for a Certificate of Correction, asking the Board to temporarily cede its exclusive jurisdiction over the proceedings.” *Id.* Thus, here, if Patent Owner wanted to correct the typographical error in the priority claim, Patent Owner should have filed a motion with the Board seeking authorization to petition the Director for a Certificate of Correction. Because it did not seek to correct the error, Patent

Owner has not established entitlement to the priority benefit of the '522 Provisional on this record.

Patent Owner then attempts to antedate the Barker reference by showing prior invention. PO Resp. 9–12. To show prior invention, Patent Owner must show either (1) conception and reduction to practice before the January 23, 2003, filing date of Barker, or (2) conception before the filing date of Barker combined with diligence and reduction to practice after that date. *See REG Synthetic Fuels, LLC v. Neste Oil Oyj*, 841 F.3d 954, 958 (Fed. Cir. 2016).

Patent Owner relies on the '522 Provisional as proof of Dr. Gierhart's conception as of December 23, 2002. PO Resp. 9–10. Patent Owner asserts that Dr. Gierhart constructively reduced the invention to practice on the same day by filing the '522 Provisional. *Id.* at 12. Alternatively, Patent Owner asserts that Dr. Gierhart was reasonably diligent from August 7, 2003, to the December 23, 2003, filing date of U.S. Application No. 10/746,403 ("the '403 Application"), the parent application of the '384 patent. PO Resp. 12 (citing Ex. 2005 ¶ 10). To show diligence, Patent Owner contends that Dr. Gierhart "continued research and development of his products and ran his new business, ZeaVision, LLC." *Id.* (citing Ex. 2005 ¶ 11).

Patent Owner's arguments fail for several reasons. First, we agree with Petitioner that the '522 Provisional was abandoned and not accorded a filing date. Ex. 1005; Ex. 1006. Accordingly, Patent Owner cannot rely on it as a constructive reduction to practice. *See In re Costello*, 717 F.2d 1346, 1350 (Fed. Cir. 1983) ("It has long been settled, and we continue to approve the rule, that an abandoned application, with which no subsequent application was copending, cannot be considered a constructive reduction to

practice.”). Here, because the ’522 Provisional did not obtain a filing date, there was no subsequent application copending, and it therefore cannot be considered a constructive reduction to practice of the claims.

Patent Owner must therefore rely on its alternative argument that Dr. Gierhart was reasonably diligent until the filing date of the ’403 Application on December 23, 2003. Patent Owner’s argument, however, fails on its face. First, as Petitioner notes, Patent Owner only asserts diligence “[f]rom prior to August 7, 2003, to the filing of [the ’403 Application] on December 23, 2003.” PO Resp. 12. Patent Owner, however, is required to show diligence starting from the period “just preceding the effective date of the adverse reference.” *See In re Steed*, 802 F.3d 1311, 1317 (Fed. Cir. 2015). That is, Patent Owner must show diligence starting from just before January 23, 2003, which is Barker’s effective filing date. Thus, Patent Owner is missing evidence of diligence for almost seven months.¹¹

Moreover, Patent Owner’s sole evidence of diligence is two paragraphs of Dr. Gierhart’s uncorroborated conclusory testimony:

10. In addition, from prior to August 7, 2003, to the filing of App. No. 10/746,403 (“Parent”), the parent application, I was reasonably continuously diligent in filing the Parent. The Parent was filed on December 23, 2003.

11. During that time, I continued research and development of my products and ran my new business, ZeaVision, LLC, devoted to development and manufacturing of the products. During that period, I engaged in reasonably continuous efforts given my business obligations towards filing

¹¹ Even if Patent Owner’s claim of diligence from “prior to August 7, 2003” were construed to extend to just before January 23, 2003, Patent Owner’s argument still fails because of the lack of corroborative evidence, as explained below.

the Parent. During that time period, my invention was not abandoned or unreasonably delayed.

Ex. 2005 ¶¶ 10–11 (cited in PO Resp. 12). Dr. Gierhart did not offer any additional evidence to support his diligence claim.

The law is clear, however, that Dr. Gierhart’s testimony, alone, is not sufficient. *Perfect Surgical Techniques, Inc. v. Olympus Am., Inc.*, 841 F.3d 1004, 1007 (Fed. Cir. 2016) (“An inventor’s testimony regarding his reasonable diligence must be corroborated by evidence.”). Here, Patent Owner has made no effort to corroborate Dr. Gierhart’s testimony, which merely parrots the Patent Owner Response. *Compare* Ex. 2005 ¶ 11, *with* PO Resp. 12.

Accordingly, having considered the arguments and evidence presented at trial, we find Patent Owner has failed to establish prior invention to Barker and reject Patent Owner’s argument that Barker is not prior art. We, therefore, move on to address the merits of Petitioner’s anticipation challenge over Barker.

b) Anticipation

Petitioner asserts that Example 6 of Barker discloses each limitation of claim 12. Pet. 41–44. We agree. Regarding the preamble, Barker discloses that colorants such as lutein and zeaxanthin absorb blue light, which can be damaging to the retina. *See* Ex. 1011, 1:10–19, 2:19–25; Ex. 1024 ¶ 115. Regarding the limitation that zeaxanthin is present in the 3R-3R' stereoisomer form and is substantially free of meso-zeaxanthin, we are persuaded that a POSITA would understand that Example 6 includes lutein and zeaxanthin, but not meso-zeaxanthin, because Barker identifies zeaxanthin and meso-zeaxanthin separately as examples of carotenoids. *See* Ex. 1011, 4–5, Example 6; Ex. 1024 ¶¶ 116–118. Regarding the

requirements for lutein, Vitamin C, and mixed tocopherols in the form of a capsule, Barker Example 6 discloses each of these ingredients in a soft gelatin capsule for oral ingestion. Ex. 1011, Example 6; Ex. 1024 ¶¶ 119–122. Specifically, for the mixed tocopherols, Barker Example 6 discloses soybean oil, which was known in the art to contain mixed tocopherols. *See* Ex. 1017, 47. Thus, we are persuaded that Petitioner has shown by a preponderance of the evidence that Barker anticipates claim 12. *See* Pet. 41–44 (citing Ex. 1011, 1:10–19, Example 6; Ex. 1024 ¶¶ 115–122).

Patent Owner argues that Barker does not disclose that zeaxanthin is present in the 3R-3R' stereoisomer form and is substantially free of meso-zeaxanthin. PO Resp. 26–27. Patent Owner asserts that although Barker discusses examples containing zeaxanthin, Barker fails to “(1) specifically disclaim the use of meso-zeaxanthin in those formulations or (2) explain the benefits of using zeaxanthin as opposed to meso-zeaxanthin.” *Id.* at 27. Patent Owner asserts that because zeaxanthin found in the eye is a mixture of the various isoforms, it would not have been immediately apparent to a POSITA that the zeaxanthin in Barker should be substantially free of meso-zeaxanthin. *Id.* (citing Ex. 2002 ¶¶ 55–56).

We are not persuaded. We disagree that Barker must either “specifically disclaim” meso-zeaxanthin or explain the benefits of using zeaxanthin over meso-zeaxanthin to disclose the limitation of “substantially free of meso-zeaxanthin.” As the Federal Circuit has held, “[a] reference need not state a feature’s absence in order to disclose a negative limitation.” *Almirall, LLC v. Amneal Pharms. LLC*, 28 F. 4th 265, 273 (Fed. Cir. 2022). Rather, the question is whether, in the context of Barker, a POSITA would have understood that the Barker formulation of Example 6 was a complete formulation that was substantially free of meso-zeaxanthin. *See id.*

(affirming the Board’s finding that the prior art reference taught the negative limitation where “a skilled artisan would recognize that the reference discloses a complete formulation—excluding the possibility of an additional active ingredient”). We find that it does.

Read as a whole, Barker consistently distinguishes zeaxanthin (i.e., the 3R-3R' stereoisomer) from meso-zeaxanthin (i.e., the 3S-3R' stereoisomer) throughout the specification. Barker repeatedly discloses “lutein, zeaxanthin, [and] mesozeaxanthin” separately. *See, e.g.*, Ex. 1011, 1:23–24, 2:6–7, 3:16–17, 36, 4:4. And certain of Barker’s claims recite lutein, zeaxanthin, and meso-zeaxanthin (*id.*, claims 4, 7, 10, 21), whereas others depend from those claims and further recite only lutein and zeaxanthin (*id.*, claims 7, 8, 24, 25). In light of the clear distinction Barker has set forth between zeaxanthin and meso-zeaxanthin, Dr. Johnson explains that a POSITA “would envisage that the zeaxanthin in Barker [Example] 6 is the 3R-3R' stereoisomer form of zeaxanthin and that it would be substantially free of meso-zeaxanthin stereoisomers.” *See* Ex. 1024 ¶¶ 116–118; *see also* Ex. 1011, 1:15–17, Example 6. In contrast, Patent Owner’s expert Dr. Dorey fails to address the disclosure of Barker as a whole and merely states in a conclusory, unsupported fashion that “[i]t would not be readily apparent to a POSITA at the time of the invention of the ’384 Patent that the zeaxanthin claimed in Barker should be substantially free of meso-zeaxanthin.” Ex. 2002 ¶ 56. We, therefore, credit the testimony of Dr. Johnson and find that a POSITA—reading Barker as a whole—would have understood that Example 6 discloses the use of zeaxanthin that is substantially free of meso-zeaxanthin, as required by claim 12. *See* Ex. 1024 ¶¶ 116–118; *see also* Ex. 1011, 1:15–17, Example 6.

Petitioner also asserts that Example 6 of Barker discloses each limitation of dependent claims 17, 18, and 20. Pet. 44–45 (citing Ex. 1011, Example 6; Ex. 1024 ¶¶ 123–124). Patent Owner does not address those dependent claims beyond its arguments regarding claim 12. PO Resp. 28; PO Sur-reply 11–12. We, therefore, determine that Petitioner has shown by a preponderance of the evidence that Barker anticipates claims 17, 18, and 20, for the reasons stated by Petitioner. *See* Pet. 44–45; Ex. 1011, Example 6; Ex. 1024 ¶¶ 123–124.

3. *Conclusion*

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 12, 17, 18, and 20 are anticipated by Barker.

E. Obviousness over Lang, Garnett, and Brown

Petitioner asserts that claims 12, 13, 16–21, 23, and 24 are unpatentable as obvious over Lang and Garnett, and that claim 14 is unpatentable as obvious over Lang, Garnett, and Brown. Pet. 45–61. Patent Owner opposes Petitioner’s assertion. PO Resp. 31–32. Having considered the arguments and evidence presented at trial, we determine that Petitioner has shown by a preponderance of the evidence that claims 12, 13, 16–21, 23, and 24 are unpatentable as obvious over Lang and Garnett, and that claim 14 is unpatentable as obvious over Lang, Garnett, and Brown. We incorporate here our earlier findings and discussion regarding the disclosure of Lang.

1. *Garnett (Ex. 1012)*

Garnett is entitled “Method of Using Pure 3R-3'R Stereoisomer of Zeaxanthin to Treat or Prevent Retinal Degeneration in Humans,” and was filed on October 31, 1995, and issued on May 5, 1998. Garnett is, therefore,

prior art under § 102(b), which Patent Owner does not dispute. *See generally* PO Resp.

Garnett relates to a method “for treating or preventing macular degeneration . . . by administering a preparation of zeaxanthin which contains the 3R-3'R stereoisomer as a sole or heavily dominant stereoisomer.” Ex. 1012, Abstract. Garnett further teaches that zeaxanthin is naturally present in macular cells and “absorbs blue and near-ultraviolet light radiation, thereby protecting retinal cells against phototoxic damage.” *Id.* Thus, according to Garnett, the purified R-R isomer of zeaxanthin can be taken orally as a nutritional supplement by patients suffering from macular degeneration or who want to reduce their risk of macular degeneration. *Id.*

2. *Brown (Ex. 1014)*

Brown is a review article entitled “Nutrition supplements and the eye,” and appears to have been published in the journal *Eye* in February 1998. Ex. 1014, 127. Brown is, therefore, prior art under § 102(b), which Patent Owner does not dispute. *See generally* PO Resp.

Brown describes “the role of vitamins, minerals, carotenoids and essential fatty acids in relation to eye health.” Ex. 1014, Abstract. Brown describes that “[t]he omega-3 fatty acids are essential for the normal development of the eye and brain in the infant.” *Id.* at 131. Brown further describes that “[o]mega-3 fatty acids are essential to retinal function at all ages and are found in the receptor outer segments.” *Id.* As further taught by Brown, “[c]onsumption of omega-3 fatty acids has been shown to reduce the incidence of cardiovascular disease, so that they can be considered to have a role of promoting general health with an indirect benefit to the eye.” *Id.*

3. *Analysis*

A patent claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which the subject matter pertains. *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations, including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of skill in the art; and (4) objective evidence of nonobviousness. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

“[A] patent composed of several elements is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR*, 550 U.S. at 418. “[I]t can be important to identify a reason that would have prompted a person of ordinary skill in the relevant field to combine the elements in the way the claimed new invention does.” *Id.* Moreover, a person of ordinary skill in the art must have had a reasonable expectation of success of doing so. *PAR Pharm., Inc. v. TWi Pharms., Inc.*, 773 F.3d 1186, 1193 (Fed. Cir. 2014).

Petitioner asserts that claims 12, 13, 17–21, 23, and 24 are unpatentable as obvious over Lang and Garnett. Pet. 47–56. Patent Owner argues that Petitioner has failed to meet its burden because it has failed to show the cited references teach each limitation of claim 12. PO Resp. 31. For the same reasons stated above, Patent Owner asserts Lang does not teach mixed tocopherols and zeaxanthin that is substantially free of meso-zeaxanthin, and that Petitioner failed to argue that Garnett and Brown

contain mixed tocopherols.¹² *Id.* Patent Owner further argues that the challenges to dependent claims 13, 14, 16–20, 23, and 24 all fall for the same reasons as claims 12, 17, and 18, discussed above. *Id.* at 32.

As explained above, we do not find Patent Owner’s arguments persuasive. Because we find that Petitioner has shown by a preponderance of the evidence that Lang anticipates claims 12, 13, 17–20, 23, and 24, we find Petitioner has also shown by a preponderance of the evidence that Lang and Garnett render those claims obvious, as it is well settled that anticipation is the epitome of obviousness.¹³ *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1373 (Fed. Cir. 2019) (“[I]t is well settled that a disclosure that anticipates under § 102 also renders the claim invalid under § 103, for anticipation is the epitome of obviousness.”) (internal quotations and citations omitted).

To the extent Petitioner further relies on Garnett’s teaching regarding protecting a user’s eyes against the harmful effect of blue light (Ex. 1012, Abstract) and regarding the highly purified 3R-3R’ stereoisomer (Ex. 1012, 16:43–48), we are persuaded that Garnett also teaches those limitations and that a person of ordinary skill in the art would have been motivated to combine the purified zeaxanthin of Garnett with the Example 12-3

¹² Patent Owner states “Petitioner failed to argue that Roche and Brown contain mixed tocopherols.” PO Resp. 31. We assume this is a typographical error, and that Patent Owner meant to say “Garnett and Brown” in this section of its brief addressing “Lang in view of Garnett and Brown.” *Id.*

¹³ We further note that there is no objective evidence of nonobviousness in the record, which Patent Owner confirmed during oral hearing. *See* Tr. 48:18–19 (Patent Owner’s counsel stating “we didn’t put on any evidence of secondary considerations”); *see also Graham*, 383 U.S. at 17–18.

formulation of Lang for the reasons stated by Petitioner. Pet. 51–53; *see also* Ex. 1024 ¶¶ 151–154. That is, we find that a person of ordinary skill in the art would have been motivated to combine Garnett and Lang because both are directed to the same problem of promoting ocular health and preventing/treating ocular disorders using formulations comprising zeaxanthin. *See* Ex. 1009, 4:7–18; Ex. 1012, Abstract; Ex. 1024 ¶¶ 151–152.

Claims 14 and 16 depend from claim 12 and further recite “omega-3 fatty acids that are derived from fish oil” and “the daily dosage of zeaxanthin is at least 10 milligrams,” respectively. Ex. 1001, 37:27–29, 38:5–6. As explained above, we find that Lang discloses each limitation of claim 12, and that the combination of Lang and Garnett renders claim 12 obvious. *See supra*. As for the additional limitation of claim 14, we agree with Petitioner that Brown teaches that omega-3 fatty acids are an essential nutrient for the eyes and can be found in fish oil. *See* Ex. 1014, 131. We also credit the testimony of Dr. Johnson that a POSITA would have been motivated to incorporate the omega-3 fatty acids of Brown into the zeaxanthin formulation of Lang with a reasonable expectation of success because both teach nutrients that are beneficial to eye health. Ex. 1024 ¶¶ 165–166.

Regarding the limitation of claim 16 that the “daily dosage of zeaxanthin is at least 10 milligrams,” we agree with Petitioner that Garnett teaches zeaxanthin formulations comprising a daily dosage of about 0.5 mg to 25 mg of 3R-3R' zeaxanthin. *See* Ex. 1012, 17:52–57. Moreover, we find that a POSITA would have been motivated to incorporate at least 10 mg and up to 25 mg of 3R-3R' zeaxanthin as taught by Garnett into the zeaxanthin formulation of Lang to protect a user's eyes from the harmful effect of blue light with a reasonable expectation of success. *See* Ex. 1024 ¶ 173.

Accordingly, we determine that Petitioner has shown by a preponderance of the evidence that each limitation of claims 14 and 16 is taught by the combination of Lang, Garnett, and Brown and that a person of ordinary skill in the art would have had a reason to combine the references with a reasonable expectation of success. *See* Ex. 1024 ¶¶ 131–132, 163–173.

Regarding claim 21, Petitioner asserts that Lang renders the claim obvious because it “expressly teaches the use of its formulations in users having macular degeneration and a lens-related disorder.” Pet. 60. Patent Owner asserts that the combination of Lang and Garnett does not teach a user who has both disorders and therefore fails to disclose all elements of claim 21. PO Resp. 32. As explained above, we are not persuaded on this record that Lang expressly teaches using the formulation in users having both disorders. But, as Petitioner notes, Lang teaches that the use of xanthophylls like lutein and zeaxanthin “may be beneficial in improving the health of the macula and the clarity of the lens.” *See* Pet. 59 (citing Ex. 1009, 2:17–19; Ex. 1024 ¶ 175). Given this suggestion that lutein and zeaxanthin may be beneficial to both the macula and the lens, we are persuaded that a person of ordinary skill in the art reading Lang would have found it obvious to use the formulation in users with both macular degeneration and a lens-related disorder. *See* Ex. 1024 ¶ 175.

4. Conclusion

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 12, 13, 16–21, 23, and 24 are unpatentable as obvious over Lang and Garnett, and that claim 14 is unpatentable as obvious over Lang, Garnett, and Brown.

F. Obviousness over Lang, Roche, and Brown

Petitioner asserts that claims 12, 13, 17–21, 23, and 24 are unpatentable as obvious over Lang and Roche. Pet. 61–76. Petitioner further asserts that claim 14 is unpatentable as obvious over Lang, Roche, and Brown. Pet. 72–74. Patent Owner opposes Petitioner’s assertion. PO Resp. 33–34. Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 12, 13, 17–21, 23, and 24 would have been obvious over Lang and Roche, and claim 14 would have been obvious over Lang, Roche, and Brown. We incorporate here our earlier findings and discussion regarding the disclosures of Lang and Brown.

1. Roche (Ex. 1020)

Roche is an FDA premarket notification of a new dietary ingredient, zeaxanthin, dated June 13, 2001. Ex. 1020, 1.¹⁴ Roche is, therefore, prior art under § 102(b), which Patent Owner does not dispute. *See generally* PO Resp.

Roche describes that zeaxanthin is a dietary ingredient for dietary supplements, which belongs to a class of carotenoids called xanthophylls. *Id.* at 14. Roche teaches that three isomers of zeaxanthin are normally present in the retina of which the most abundant isomer is “(3R,3’R)-zeaxanthin.” *Id.* Roche also states that “[t]he presently described dietary ingredient contains almost exclusively (3R,3’R)-zeaxanthin, as well as very small amounts of (3R,3’S)-(meso)-zeaxanthin and (3S,3’S)-zeaxanthin.” *Id.* Roche states that the “product will be available as a bulk ingredient in a

¹⁴ Citations to Roche refer to the page numbers of the document and not the page numbers added to the lower right corner of the page.

beadlet formulation containing 5% zeaxanthin (Zeaxanthin 5% TG), for the manufacture of dietary supplements.” *Id.* at 12.

2. *Analysis*

Petitioner asserts that the combination of Lang and Roche teaches each limitation of claims 12, 13, 17–21, 23, and 24. Pet. 61–76. Because we find above that Petitioner has shown by a preponderance of the evidence that Lang anticipates claims 12, 13, 17–20, 23, and 24, we find Petitioner has also shown by a preponderance of the evidence that Lang and Roche render those claims obvious, as well. *See Realtime Data, LLC*, 912 F.3d at 1373. To the extent Petitioner further relies on Roche’s teaching regarding protecting a user’s eyes against the harmful effect of blue light (Ex. 1020, 23–33) and regarding the highly purified 3R-3R’ stereoisomer (Ex. 1020, 14), we are persuaded that Roche also teaches those limitations and that a person of ordinary skill in the art would have been motivated to combine Roche’s zeaxanthin 5% TG beadlet formulation, which includes very little meso-zeaxanthin with the Example 12-3 formulation of Lang, for the reasons stated by Petitioner. Pet. 67–69; Ex. 1024 ¶¶ 205–209. For example, we credit the testimony of Dr. Johnson and find that a POSITA would have been motivated to combine Lang and Roche because both are directed to the same problem of “promoting ocular health and preventing/treating ocular disorders, such as macular degeneration, using formulations comprising zeaxanthin.” *See* Ex. 1024 ¶ 205; *see also* Ex. 1009, 4:7–18; Ex. 1020, 17–33.

For the same reasons stated above, we find that Brown teaches the “omega-3 fatty acid” limitation of claim 14 and that a person of ordinary skill in the art would have been motivated to combine the omega-3 fatty

acids of Brown with the zeaxanthin formulation of Lang with a reasonable expectation of success. *See supra*.

Finally, Petitioner asserts that claim 21 would have been obvious over Lang and Roche. Pet. 74–75. Patent Owner argues that Lang, Roche, and Brown fail to disclose all elements of claim 21. PO Resp. 33. Specifically, Patent Owner argues that Brown suggests that the use of vitamin supplements is unlikely to be beneficial to lens-related disorders. *Id.* at 34. Because of that, Patent Owner argues a POSITA would not have been motivated to combine Brown and Lang with a reasonable expectation of success. *Id.* Moreover, Patent Owner argues that Roche does not teach the use of 3R-3R' zeaxanthin for preventing or treating macular degeneration and lens-related disorders. *Id.*

For the same reasons stated above with respect to Lang and Garnett, we are persuaded that Petitioner has shown by a preponderance of the evidence that Lang suggests the use of the formulation in users with both macular degeneration and lens-related disorders, as required by claim 21. *See supra*. To the extent Patent Owner argues a POSITA would not have been motivated to combine Brown and Lang with a reasonable expectation of success to render claim 21 obvious (PO Resp. 34), Petitioner did not rely on Brown for its challenge. *See* Pet. 74–75. Thus, we agree that Patent Owner's arguments are inapposite.

3. Conclusion

Having considered the arguments and evidence presented at trial, we find Petitioner has shown by a preponderance of the evidence that claims 12, 13, 17–21, 23, and 24 are unpatentable as obvious over Lang and Roche, and claim 14 would have been unpatentable as obvious over Lang, Roche, and Brown.

III. CONCLUSION¹⁵

For the foregoing reasons, we determine that Petitioner has established by a preponderance of the evidence that claims 12–14, 16–21, 23, and 24 of the '384 patent are unpatentable.

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not shown Unpatentable
12, 13, 17–21, 23, 24	102(b)	Lang	12, 13, 17–20, 23, 24	21
12, 17, 18, 20	102(a), (e)	Barker	12, 17, 18, 20	
12, 13, 16–21, 23, 24	103(a)	Lang, Garnett	12, 13, 16–21, 23, 24	
14	103(a)	Lang, Garnett, Brown	14	
12, 13, 17–21, 23, 24	103(a)	Lang, Roche	12, 13, 17–21, 23, 24	
14	103(a)	Lang, Roche, Brown	14	
Overall Outcome			12–14, 16–21, 23, 24	

¹⁵ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 12–14, 16–21, 23, and 24 of U.S. Patent No. 10,307,384 B2 are held unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, the parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirement of 37 C.F.R. § 90.2.

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